IEEE Access
Multidisciplinary : Rapid Review : Open Access Journal

Received 23 October 2025, accepted 14 December 2025, date of publication 25 December 2025, date of current version 22 January 2026.

Digital Object Identifier 10.1109/ACCESS.2025.3648467

RESEARCH ARTICLE

From Time Series to Images: A GAF-ResNet-LSTM Framework for Asset Selection and Portfolio Optimization

NEHA SOLANKI AND MANOJ JHA
Department of Mathematics, Bioinformatics, and Computer Applications, Maulana Azad National Institute of Technology, Bhopal, Madhya Pradesh 462003, India
Corresponding author: Neha Solanki (223130006@stu.manit.ac.in)

ABSTRACT Financial market forecasting in portfolio formation has increasingly drawn interest within machine learning due to its complex temporal dynamics and high volatility. While Convolutional Neural Networks (CNN) have demonstrated efficacy in image-based tasks, their direct application to time series data remains limited. To address this, we propose a prediction-based portfolio optimization framework that converts stock price sequences into Gramian Angular Field (GAF) images, enabling spatial feature extraction via a ResNet. Temporal dependencies are captured using a long short term memory (LSTM) encoder, forming a hybrid GAF + ResNet-LSTM model. To be specific, this paper first applies these prediction models for asset preselection before portfolio construction. Then, this paper incorporates their predictive results in advancing nested clustered optimization (NCO) method, which clusters assets based on predictive confidence and allocates weights using intra-cluster optimization. In order to balance calibration and classification, training employs walk-forward validation and a composite loss combining mean squared error and directional accuracy. The empirical study is conducted on a dataset comprising 25 stocks from the National Stock Exchange of India, spanning January 2014 to January 2024, including periods of market disruption such as the COVID-19 crash. Thus the sample makes it possible to catch subtle market shift for model execution. Moreover, we compared the performance of the proposed model against equally weighted and mean-variance portfolios to underscore the advantages of the NCO model. The proposed model achieves a 21.05% annual return with a Sharpe ratio of 1.31, outperforming benchmark architectures. Empirical results demonstrate that the proposed technique significantly outperforms the benchmark portfolios, particularly for investors with a high risk-return preference. Statistical significance is further corroborated by t-tests. Based on these findings, we endorse the integration of our comprehensive portfolio optimization model into investment decision-making processes.

INDEX TERMS Financial forecasting, GAF, LSTM, NCO, portfolio optimization, residual network.

I. INTRODUCTION

Portfolio selection aims to construct an asset allocation that optimally balances expected returns against associated risks. Under the classical mean-variance framework, this trade-off is formalized through the efficient frontier, which identifies portfolios that maximize expected return for a given level of risk, with the final choice determined by an investor's risk preferences [1], [2]. Markowitz's seminal work [3] established that diversification across imperfectly correlated assets can significantly reduce portfolio risk, forming the foundation of modern portfolio theory. However, despite its theoretical elegance, the practical success of portfolio optimization remains critically dependent on the quality of its inputs- the selection of assets and the accuracy of expected return forecasts [4], [5]. Accurate return forecasting is widely acknowledged as challenging due to the nonlinear, noisy, and chaotic nature of financial markets. Tradi-

The associate editor coordinating the review of this manuscript and approving it for publication was Chuan Li.

tional statistical models such as autoregressive integrated moving average (ARIMA) and generalized autoregressive conditional heteroscedasticity (GARCH), as well as early machine learning approaches including recurrent neural networks (RNN), have been widely applied but often fail to capture the full complexity of financial time series [6], [7], [8], [9]. Although more advanced neural architectures have improved predictive performance, return estimation remains a major bottleneck in achieving robust out-of-sample portfolio performance. Consequently, recent studies increasingly emphasize prediction-based portfolio optimization, where improved forecasting models are leveraged to construct superior efficient frontiers and reduce realized risk.

In response, a growing body of literature has explored diverse methodological innovations. Reinforcement learning-based approaches dynamically adjust portfolio weights in response to real-time market feedback [10], while graph-theoretic models explicitly encode inter-asset dependencies to enhance diversification benefits [11]. Other studies adopt credibility theory and uncertainty-aware frameworks to address estimation risk when historical data are sparse or unreliable [2]. While these methods contribute valuable insights, they largely rely on numerical return representations and often treat forecasting and allocation as loosely coupled stages, limiting their ability to exploit complex temporal structures embedded in financial data. From a methodological perspective, contemporary machine learning-driven portfolio optimization strategies can be broadly categorized into three groups: (i) machine learning–based stock preselection followed by classical optimization [12], [13], [14]; (ii) direct incorporation of predicted returns into portfolio objective functions [15], [16]; and (iii) substitution of historical returns with prediction errors to exploit desirable statistical properties such as normality [12], [14], [17]. Although these approaches demonstrate incremental improvements, they are typically developed in isolation and do not fully exploit the synergistic potential of advanced representation learning and sequential modeling. A significant methodological transition has emerged in transforming time-series data into image representations to utilize the advanced feature extraction capabilities of convolutional neural networks [18]. Techniques such as recurrence plots (RP) and Gramian Angular Fields (GAF) convert one-dimensional temporal sequences into two-dimensional texture-like images, enabling forecasting to be framed as a visual pattern recognition problem. Hatami et al. [19] demonstrated that RP-based image encodings combined with CNNs can outperform traditional approaches in time-series classification tasks. Building on this idea, Barra et al. [20] employed GAF-based representations for financial forecasting and showed that CNN ensembles trained on such images can outperform conventional investment strategies such as buy-and-hold. Despite these promising results, existing image-based approaches primarily focus on single-scale representations and prediction accuracy, with limited consideration of multi-horizon dynamics and portfolio-level decision requirements.

This study addresses these limitations by proposing a unified prediction-based portfolio optimization framework that advances both deep learning and portfolio optimization research. Specifically, we develop a hybrid model that integrates GAF-based image representations with deep learning architectures to capture nonlinear and multi-scale temporal patterns in financial time series. Unlike prior work that treats forecasting and allocation as separate components, the proposed approach is explicitly designed to generate forecasts that are structurally aligned with portfolio optimization objectives, thereby enhancing out-of-sample performance under comparable risk levels. The main objective of this study is to construct a more efficient prediction-based portfolio optimization model that delivers improved returns without increasing risk, while maintaining robustness in out-of-sample regimes.

Recent advances in deep learning have significantly expanded the methodological toolkit for financial forecasting and portfolio optimization. Akita et al. [21] showed early on that multivariate LSTM architectures trained on cross-sectional equity data do better than univariate models because they explicitly capture inter-stock dependencies. The introduction of residual learning by He et al. [22] alleviated vanishing-gradient limitations and enabled the training of deeper and more expressive neural networks, which subsequently motivated the adoption of advanced architectures in financial modeling. Fischer and Krauss [4] showed that LSTM-based strategies can effectively capture temporal dependencies in equity returns, achieving economically meaningful excess returns on S&P 500 stocks. Complementing this line of research, Sezer and Ozbayoglu [23] proposed CNN-TA, a two-dimensional convolutional framework that encodes technical indicators as image channels, outperforming traditional rule-based trading systems. The use of visual representations has further broadened the scope of deep learning applications in finance. Komori [24] employed Inception-v3 networks trained on candlestick chart images to model price movements, while Wang [14] adapted ResNet architectures for financial time series, improving training stability and gradient propagation through skip connections. These studies collectively point out the value of image-based and deep residual architectures for extracting nonlinear patterns from market data.

Despite these advances, several limitations remain. Kim and Kim [25] reported improved trading profitability using CNN-LSTM architectures applied to stock chart images, but their approach was criticized for insufficiently modeling sequential dependencies. Lee and Kim [26] addressed the issue by applying CNN-LSTM models to high-dimensional time series, achieving substantially higher cumulative returns than buy-and-hold strategies, yet their performance deteriorated under regime shifts. These findings point out the importance of architectures that can adaptively balance spatial and temporal feature extraction while remaining robust across diverse market conditions. More recent research has therefore shifted toward hybrid and robust modeling frameworks. Costa and Kwon [27] integrated ResNet-LSTM architectures with optimization pipelines that explicitly control portfolio-level risk and marginal risk contributions, achieving improved diversification and risk-adjusted performance. López de Prado emphasized the role of machine learning in uncovering complex nonlinear interactions that are difficult to capture using classical econometric approaches [28]. Building on this perspective, Imajo et al. introduced residual networks with adaptive attention mechanisms to enhance robustness in volatile markets [29], while Khodaee [30]. proposed a ResNet-CNN-LSTM hybrid that combines hierarchical feature extraction with long-term dependency modeling Du [31] further demonstrated that integrating ARIMA-based forecasts into mean-variance and omega optimization frameworks can yield superior risk-adjusted returns compared to standalone strategies. More recently, Singh et al. [32] incorporated both technical and fundamental features within CNN-LSTM architectures for stock selection, outperforming single-model baselines, while Peng et al. [34] fused CNN with attention-based LSTM for volatility prediction, achieving notable accuracy gains. These developments reflect a growing consensus that hybrid architectures are essential for capturing the complex spatial-temporal dynamics of financial markets. In parallel with advances in forecasting models, portfolio optimization research has evolved to better account for uncertainty, market heterogeneity, and investor preferences. Recent empirical studies applying the classical mean-variance framework to emerging markets highlight its continued relevance while also emphasizing sensitivity to return estimation error and market-specific dynamics [35]. Multi-objective credibilistic portfolio optimization models extend traditional formulations by jointly considering return, risk, and liquidity, offering greater flexibility in both emerging and developed markets [36]. Other studies propose uncertainty-aware formulations, such as mean-chance portfolio models that explicitly incorporate background risk, thereby improving robustness under stochastic environments [37]. Behavioral and decision-theoretic extensions, including fuzzy multi-criteria portfolio selection based on three-way decisions and cumulative prospect theory, further enrich the theoretical paradigm by capturing ambiguity and investor psychology [38]. Clustering-based and adaptive optimization techniques have also gained prominence. Nested Clustered Optimization, introduced by López de Prado [39], groups assets according to similarity in risk-return characteristics prior to optimization, reducing overfitting and improving out-of-sample performance. Espiga-Fernández et al. [40] combined spectral clustering with reinforcement learning for dynamic portfolio rebalancing, achieving higher Sharpe ratios than traditional mean-variance benchmarks, while Lemieux et al. [41] demonstrated the advantages of clustering-based strategies for risk-averse investors managing high-dimensional portfolios. At the same time, Sharma et al. [42] cautioned against over-reliance on predictive accuracy alone and emphasized the importance of ablation studies to isolate model contributions. Hu and Kercheval, as well as Zhang et al. [43], [44], stressed the necessity of rigorous statistical validation-such as bootstrapping and hypothesis testing-when claiming superior performance of deep learning-driven portfolio strategies. These changes show that more and more people agree that building a good equity portfolio requires both advanced forecasting systems and strong, data-driven optimization models.

A. KEY CONTRIBUTIONS

This study advances financial time-series forecasting and portfolio optimization in several ways.

- We introduced a prediction model that converts price sequences into GAF images for spatial feature extraction via a ResNet and aggregates slice-level representations with an LSTM to capture temporal dependencies. This design jointly exploits local shape motifs and cross-segment dynamics.
- We employed a multi-resolution imaging pipeline that preserves fine- and coarse-grained structure in price evolution, improving class-balanced recognition like *Down* regimes and enabling more effective convolutional feature learning.
- A composite loss combining mean-squared error and directional accuracy is used during walk-forward training to balance probabilistic calibration with sign prediction.
- We adopted a fixed, non-overlapping walk-forward validation technique from 2014 to 2024 that respects temporal causality and spans multiple market regimes, including COVID-19, yielding realistic out-of-sample assessments.
- Forecasts are coupled with Nested Clustered Optimization that groups economically coherent assets (Pearson-Ward clustering) and performs constraint-aware intra-cluster allocation, enhancing diversification and estimation stability.
- Relative to equal-weighted and mean-variance baselines, as well as a recent deep learning benchmark, the proposed framework achieves the highest cumulative return and Sharpe ratio, the lowest maximum drawdown (-18.2%), and competitive volatility, evidencing meaningful risk-adjusted gains.
- Performance differences are evaluated with paired t-tests on aligned return series; results reject equality of means in favor of the proposed approach, supporting that improvements are not due to sampling variability.
- We document all preprocessing, walk splits, and hyperparameters, and report a consistent suite of classification and portfolio metrics (accuracy, macro/weighted-F1,

Sharpe, volatility, and drawdown) to facilitate replication and fair comparison.

II. METHODOLOGY

In the proposed study, financial time series are sequentially transformed into image representations and subsequently modeled to predict return directions of assets. The pipeline begins with data collection and preprocessing, after which the raw sequences are encoded into GAF images at multiple resolutions to capture both local and global structural patterns. These representations are then processed through a hybrid ResNet-LSTM predictor, where convolutional layers extract spatial dependencies from the GAF images and recurrent layers capture temporal dynamics inherent in financial data. The predictive outputs are further integrated into a portfolio construction strategy using NCO, which exploits probabilistic signals to enhance diversification and risk-adjusted returns. Model performance is rigorously assessed through a walk-forward evaluation technique designed to mimic realistic investment workflows. A detailed overview of these sequential steps is provided in Figure. 1.

A. DATA COLLECTION

The empirical analysis in this study is based on daily stock data obtained from the National Stock Exchange (NSE) of India. A cross-section of 25 equities was chosen to guarantee sufficient sectoral representation and market diversity. The dataset covers a ten-year period from January 2014 to January 2024, which includes both stable and unstable market conditions. The COVID-19 crisis of 2019-2021, for example, showed how much more uncertain and risky financial markets can be. For each stock, trading days with reliable open and close prices were retained, while duplicated or erroneous time stamps were systematically removed. Adjusted closing prices were used as the primary input to account for corporate actions such as dividends and stock splits. In cases where adjusted prices were unavailable, standard closing prices were substituted following quality control checks. To ensure comparability across securities, all time series were aligned on common trading days, resulting in a uniformly structured panel dataset. The complete list of stock symbols included in the study is reported in Table 1. The selected ten-year horizon is consistent with prior empirical studies in financial forecasting and portfolio optimization [14], [26], [33] and is sufficiently long to yield statistically robust insights while capturing multiple market regimes.

B. GRAMIAN ANGULAR FIELD ENCODING

Gramian Angular Field (GAF) encoding converts one-dimensional time-series data into two-dimensional image representations, enabling the use of convolutional neural networks (CNN) for financial forecasting. Originally proposed by Wang and Oates [18], GAF is particularly attractive in our setting because it preserves temporal ordering while yielding a spatial structure that CNN backbones (including pre-trained vision models) can exploit.

Given a univariate time series $X = \{x_1, x_2, \ldots, x_N\}$, we first normalize the sequence to the interval $[-1, 1]$ using min–max scaling:

$$\tilde{x}_i = \frac{(x_i - \max(X)) + (x_i - \min(X))}{\max(X) - \min(X)}, \quad i = 1, \ldots, N.$$

This step improves numerical stability and ensures compatibility with the angular mapping that follows. Each normalized value is mapped to polar coordinates:

$$\phi_i = \arccos(\tilde{x}_i), \quad r_i = \frac{i}{N}.$$

Here, ϕ_i encodes the magnitude information as an angle, while r_i encodes the temporal index, which preserves the chronological structure in the resulting image.

Gramian Difference Angular Field (GADF): Among GAF variants, we employ the GADF because it emphasizes directional changes and transition dynamics such as momentum and reversal patterns that are central to return forecasting. The GADF matrix is defined as

$$\mathrm{GADF}_{i,j} = \sin(\phi_i - \phi_j), \quad i, j = 1, \ldots, N.$$

The resulting matrix interpreted as a grayscale image with values in $[-1, 1]$. For compatibility with standard CNN backbones and to enhance contrast during training, we apply a colormap to obtain a three-channel (RGB) representation; this procedure is a visualization-to-learning convenience and does not change the underlying encoded structure.

C. RESIDUAL NEURAL NETWORKS

He et al. [22] introduced a deep residual network that tackles the gradient vanishing issue by employing skip connections, which allow information to bypass intermediate layers. In ResNet, a layer's output is passed directly to the layer two or three steps ahead, which also accelerates the training process. To clarify this concept, Figure. 2 illustrates a residual block made up of three weight layers, and below equation details the forward propagation computation in a ResNet-equipped network.

$$a^l = g\left(w^{l-1,l} \cdot a^{l-1} + b^l + w^{l-2,l} \cdot a^{l-2}\right)$$

where a^l is the output of layer l, b^l is the bias, g is the activation function, such as ReLU, for the layer of l, and $w^{l-1,l}$ is the weight matrix between the layers of $l-1$ and l. Note that in the absence of $w^{l-2,l}$, especially in the initializing step, the identity matrix can be used [30].

ResNet learns residual functions, where each block computes the difference between the input and desired output rather than the output itself. This formulation allows gradient flow to bypass certain layers, improving convergence and stability. The provided implementation includes two key components: identity blocks, which maintain input dimensions, and convolutional blocks, which adjust dimensions using strided convolutions and projection shortcuts. These residual blocks ensure efficient feature extraction while preserving critical information. In the given ResNet-based

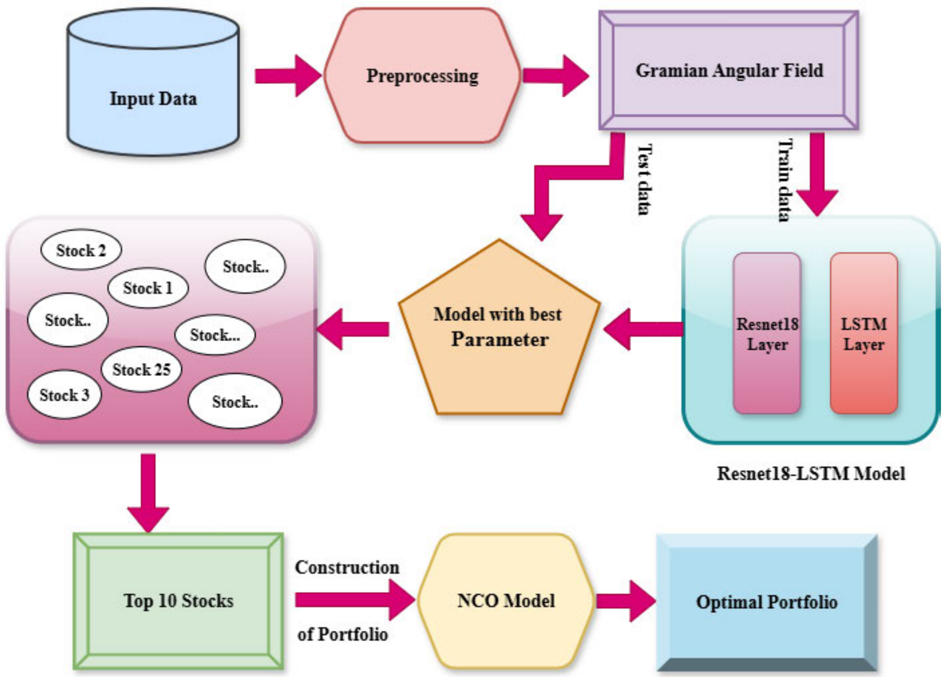

FIGURE 1. Work flow of portfolio optimization.

TABLE 1. Stock exchange code of companies [45].

S.No.	Stocks	S.No.	Stocks	S.No.	Stocks	S.No.	Stocks	S.No.	Stocks
1	Adani Ports (532921)	6	Bharat Petroleum Corporation Ltd (500547)	11	Dr. Reddy's Laboratories Ltd (500124)	16	ITC Ltd (500875)	21	NTPC Ltd (532555)
2	Apollo Hospitals (508869)	7	Bharti Airtel Ltd (532454)	12	HDFC Bank Ltd (500180)	17	Infosys Ltd (500209)	22	Nestle India Ltd (500790)
3	Asian Paints (500820)	8	Cipla Ltd (500087)	13	Hero MotoCorp Ltd (500182)	18	JSW Steel Ltd (500228)	23	ONGC Ltd (500312)
4	Axis Bank (532215)	9	Coal India Ltd (533278)	14	Hindustan Unilever Ltd (500696)	19	Kotak Mahindra Bank Ltd (500247)	24	Power Grid Corporation of India Ltd (532898)
5	Bajaj Finance (500034)	10	Divi's Laboratories Ltd (532488)	15	ICICI Bank Ltd (532174)	20	Maruti Suzuki India Ltd (532500)	25	Reliance Industries Ltd (500325)

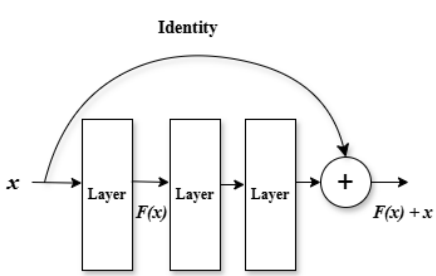

FIGURE 2. Residual block.

stock price prediction model, an initial convolutional layer extracts low-level features, followed by stacked residual blocks that capture complex dependencies in financial time series. The model employs batch normalization (BN) and ReLU activation to enhance learning efficiency. Finally, a fully connected dense layer processes the extracted representations for final prediction.

D. LONG SHORT-TERM MEMORY

LSTM, a subset of recurrent neural networks, was introduced by Hochreiter and Schmidhuber in 1997 to address the gradient vanishing problem [46]. The original LSTM architecture consisted of input gates, cells, and output gates but lacked a forget gate. Gers et al. [47] introduced the forget gate concept, enabling the network to selectively reset or forget certain states, making LSTM more suitable for sequential and continuous tasks, such as embedded Reber grammar [48]. LSTM has four components: forget gate, input gate, output gate, and cell state. LSTM controls information discarded or added through three gates, namely, the forget gate, input gate, and output gate, so as to realize the forgetting or memory function. The forget gate is a sigmoid function that controls the forgetting degree of the cell state of the previous cell through the output h_{t-1} of the previous cell and the input x_t of this cell.The input gate utilizes a tanh activation to regulate incoming information. In particular, the tanh function produces the candidate input $\bar{C}_t$, while the gate mechanism, similar to the forget gate, modulates the amount of this information integrated into the cell state. The output gate, influenced by the variable o_t and a tanh function, governs the output from the current cell. The cell state C_t is updated based on the combined effects of the forget gate. f_t and the input gate i_t.

$$lstm_{cell} = (f,i,o)$$

The respective procedures involve equations (1) to (6)

$$f_t = \sigma\ (W_f x_t + U_f h_{t-1} + b_f) \tag{1}$$
$$C_t = f_t \cdot C_{t-1} + i_t \cdot \bar{C}_t \tag{2}$$
$$i_t = \sigma(W_i x_i + U_i h_{t-1} + b_i) \tag{3}$$
$$\bar{C}_t = \tanh\ (W_C x_t + U_C h_{t-1} + b_C) \tag{4}$$
$$o_t = \sigma\ (W_o x_t + U_o h_{t-1} + b_o) \tag{5}$$
$$h_t = o_t \cdot \tanh\ (C_t) \tag{6}$$

Through the above elaborate design, the LSTM network can effectively capture the correlation between data with long intervals in time series, enabling more accurate predictions of future stock prices and fluctuations.

E. NESTED CLUSTERED OPTIMIZATION

Lopez de Prado [39] proposed the NCO algorithm. It is an advanced approach for portfolio optimization that improves the robustness and scalability of asset allocation by incorporating preselection and clustering into the optimization process. NCO addresses key challenges of traditional optimization methods, such as estimation errors in covariance matrices and computational inefficiency for large portfolios [28].

Consider N random variables with expected returns $\boldsymbol{\mu} \in \mathbb{R}^N$ and covariance matrix $\mathbf{V} \in \mathbb{R}^{N\times N}$. Let $\mathbf{a} \in \mathbb{R}^N$ be a target-return vector. The general quadratic program is

$$\min_{\boldsymbol{\omega}} \frac{1}{2}\boldsymbol{\omega}^\top \mathbf{V}\boldsymbol{\omega} \quad \text{s.t.} \quad \boldsymbol{\omega}^\top \mathbf{a} = 1$$

1) CLUSTERING

Partition the index set $\{1,\ldots,N\}$ into K disjoint clusters $\mathcal{C}_1,\ldots,\mathcal{C}_K$

$$\bigcup_{k=1}^{K} \mathcal{C}_k = \{1,\ldots,N\}, \quad \mathcal{C}_i \cap \mathcal{C}_j = \emptyset \quad (i \neq j)$$

2) INTRA-CLUSTER OPTIMIZATION

For each cluster k, let $\mathbf{V}^{(k)}$ be the $|\mathcal{C}_k| \times |\mathcal{C}_k|$ submatrix of $\mathbf{V}$ and $\mathbf{a}^{(k)}$ the corresponding subvector. The intra-cluster weight vector is

$$\boldsymbol{u}^{(k)} = \frac{(\mathbf{V}^{(k)})^{-1}\,\mathbf{a}^{(k)}}{(\mathbf{a}^{(k)})^\top(\mathbf{V}^{(k)})^{-1}\,\mathbf{a}^{(k)}} \quad k = 1,\ldots,K$$

3) COLLAPSE TO CLUSTER-LEVEL

Define the cluster-level return and covariance by

$$\tilde{\mu}_k = (\mathbf{a}^{(k)})^\top\,\boldsymbol{u}^{(k)} \quad k = 1,\ldots,K,$$
$$\tilde{V}_{ij} = (\boldsymbol{u}^{(i)})^\top\,\mathbf{V}^{(i,j)}\,\boldsymbol{u}^{(j)} \quad i,j = 1,\ldots,K$$

so that

$$\tilde{\boldsymbol{\mu}} = \left(\tilde{\mu}_1,\ldots,\tilde{\mu}_K\right)^\top \qquad \tilde{\mathbf{V}} = [\tilde{V}_{ij}]_{i,j=1}^{K}$$

4) INTER-CLUSTER OPTIMIZATION

Solve the reduced problem

$$\min_{\boldsymbol{w}\in\mathbb{R}^K} \frac{1}{2}\boldsymbol{w}^\top\tilde{\mathbf{V}}\boldsymbol{w} \quad \text{s.t.} \quad \boldsymbol{w}^\top\tilde{\boldsymbol{\mu}} = 1$$

whose closed-form solution is

$$\boldsymbol{w}^* = \frac{\tilde{\mathbf{V}}^{-1}\,\tilde{\boldsymbol{\mu}}}{\tilde{\boldsymbol{\mu}}^\top\tilde{\mathbf{V}}^{-1}\,\tilde{\boldsymbol{\mu}}}$$

5) RECONSTRUCTION OF FULL WEIGHTS

Embed the cluster-level weights back into $\mathbb{R}^N$

$$\hat{\boldsymbol{\omega}}_{\text{NCO}} = \begin{bmatrix} w_1^*\,\boldsymbol{u}^{(1)} \\ w_2^*\,\boldsymbol{u}^{(2)} \\ \vdots \\ w_K^*\,\boldsymbol{u}^{(K)} \end{bmatrix}$$

The detailed pseudocode of the proposed model is summarized in Algorithm 1.

F. ASSET SELECTION: GAF-ResNet-LSTM

Each segment $\mathbf{c}_{i,t}$ is first transformed into a fixed-size image representation using the GAF encoding. Raw prices are converted into simple returns $r_j = (p_j - p_{j-1})/p_{j-1}$, followed by a short moving average with a window size of three to suppress high-frequency noise while preserving local temporal patterns. The smoothed return series is then min–max normalized to $[0, 1]$ and uniformly resampled to a fixed length $M = 20$, ensuring consistent input dimensionality across segments. Based on this standardized sequence, the GADF transform is applied to generate an image of size $M \times M$. Each segment is thus represented as a single-channel GADF image, and stacking the T segments produces a tensor of shape $(T, 1, M, M)$. This representation encodes temporal dynamics as oriented texture patterns while providing a uniform geometric structure suitable for convolutional feature extraction. Such image-based encoding enables the model to exploit spatial inductive biases while preserving the temporal ordering of the original time series.

To jointly model intra-segment structure and inter-segment temporal dependencies, we employ a hybrid ResNet-LSTM architecture. The ResNet backbone extracts spatial features from individual GADF images, while the LSTM encoder captures sequential relationships across segments. ResNet-18 is selected due to its training stability and ability to learn hierarchical representations without degradation as network depth increases. The backbone is initialized with ImageNet pre-trained weights, and early convolutional layers are frozen during training. This design choice reflects two considerations: early layers capture generic low-level patterns transferable to GAF textures, and freezing them reduces over-fitting risk and computational cost given the moderate size of the financial dataset. Consequently, only the deeper residual blocks are fine-tuned to enable task-specific adaptation. Global average pooling is applied to the final convolutional feature maps, yielding a 512-dimensional embedding for each segment. The resulting sequence of embeddings is batch-normalized, augmented with sinusoidal positional encodings, and processed by a two-layer LSTM encoder with a hidden dimension of 512. A fixed-length sequence representation is obtained by averaging the encoded segment outputs. This transfer-learning strategy improves training stability and supports efficient walk-forward validation. An overview of the complete asset selection pipeline is illustrated in Figure. 9.

G. PROCESS OF OPTIMAL PORTFOLIO FORMATION

The second step involves identifying the portfolio that demonstrates the most favorable risk-adjusted performance and efficient capital allocation across assets. Investors are therefore expected to select portfolios that either maintain lower risk levels with consistent returns or embrace higher risk levels with the potential for increased gains. At this stage, the NCO model is applied to optimize the asset selection process. It is important to emphasize that the proposed model excludes individual investors' risk preferences and the existence of a risk-free asset; consequently, all constructed portfolios consist solely of high-risk investment options. To assess the performance and efficiency of the proposed risk-return portfolios, we employ multiple evaluation indicators, namely annual return, annual volatility, Sharpe ratio, and maximum drawdown computed from the realized daily portfolio return $\{r_t^{(p)}\}$. With $'A'$ trading days per year (here $A = 252$), define the cumulative wealth process $C_t = \prod_{\tau=1}^{t}(1 + r_\tau^{(p)})$ and its running peak $P_t = \max_{\tau \leq t} C_\tau$. The evaluation metrics are

$$\text{Annual Return} = \left(\prod_{t=1}^{T}(1 + r_t^{(p)})\right)^{A/T} - 1$$

$$\text{Annual Volatility} = \sqrt{A} \cdot \text{Std}\left(r_t^{(p)}\right)$$

$$\text{Sharpe Ratio} = \frac{\sqrt{A} \cdot \left(\overline{r_t^{(p)}} - r_f/A\right)}{\text{Std}\left(r_t^{(p)}\right)}$$

$$\text{Maximum Drawdown} = \max_{1 \leq t \leq T}\left(1 - \frac{C_t}{P_t}\right)$$

where $\overline{r_t^{(p)}}$ is the sample mean of daily portfolio returns and r_f is the annual risk-free rate. Portfolios with superior outcomes in these metrics, particularly those exhibiting stable annual returns, controlled volatility, an enhanced sharpe ratio, and a minimized drawdown, are prioritized for resource allocation. Hence, when asset weights and allocations are updated at the beginning of the subsequent trading day, capital can be effectively distributed.

H. EVALUATION METRICS

Predictions are mapped back to prices via $\hat{p}_{t+k} = p_t^{\text{ref}} \exp(\hat{y}_{t,k})$. We compute Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). Averaging across folds yields a robust estimate of expected out-of-sample error at each horizon.To assess trading relevance, we convert realized and predicted returns to three classes using a symmetric dead zone $\varepsilon = 0.5\%$

$$\text{Down if } r < -\varepsilon, \quad \text{Neutral if } |r| \leq \varepsilon, \quad \text{Up if } r > \varepsilon$$

For each horizon we report accuracy, macro-averaged precision, recall, and F1-score across {Down, Neutral, Up} using continuous scores derived from the predicted return (Up: r, Down: $-r$, Neutral: $-|r|$). Confusion matrices are saved per fold to visualize error modes; training and validation loss curves are logged to verify convergence.

1) ACCURACY

Accuracy quantifies the overall proportion of correctly classified instances relative to the total number of observations. It is formally expressed as

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

where TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives, respectively. Although accuracy is intuitive, it may provide biased assessments when class distributions are skewed.

2) PRECISION

Precision evaluates the fraction of correctly predicted positive samples among all samples predicted as positive

$$\text{Precision} = \frac{TP}{TP + FP}$$

This metric emphasizes the reliability of positive predictions, which is crucial in financial decision-making scenarios where false alarms can be costly.

3) RECALL

Recall, also referred to as sensitivity, measures the proportion of actual positive samples that are correctly classified

$$\text{Recall} = \frac{TP}{TP + FN}$$

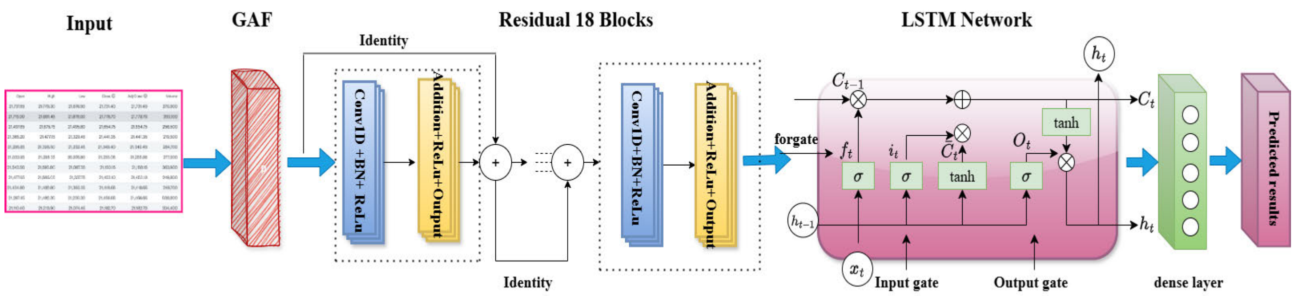

FIGURE 3. **Hybrid ResNet-LSTM model.**

It highlights the ability of the model to capture true signals and minimize missed opportunities.

4) MACRO F1

The F1 score, defined as the harmonic mean of precision and recall, is given by:

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

The Macro F1 aggregates class-wise F1 scores by taking their unweighted mean

$$\text{Macro F1} = \frac{1}{C} \sum_{i=1}^{C} F1_i$$

where C is the number of classes. This approach assigns equal importance to each class, making it suitable for evaluating performance across all categories regardless of their frequencies.

5) WEIGHTED F1

In contrast, the Weighted F1 accounts for class imbalance by weighting each class-specific F1 score according to its relative support

$$\text{Weighted F1} = \frac{1}{N} \sum_{i=1}^{C} n_i \cdot F1_i$$

where n_i is the number of true samples in class i, and N is the total number of samples. This ensures that the performance evaluation reflects the actual distribution of financial market data.

Together, these metrics provide a holistic evaluation of model performance, balancing overall accuracy, predictive reliability, sensitivity to true signals, and robustness to class imbalance.

I. BASELINE STRATEGIES FOR PORTFOLIO FORMATION

- GAF-ResNet-LSTM + MV
  The structure of this baseline model is similar to the proposed GAF-ResNet-LSTM + NCO approach in its first stage; that is, forecasting returns at time t+1 using the GAF-ResNet-LSTM architecture. However, it differs in the second stage, where Markowitz's mean-variance optimization is employed instead of the NCO approach. Specifically, after predicting future returns, those assets with higher forecasted returns are selected. The total number of selected assets matches in the proposed model. After that, in the second stage, the classical MV framework is applied to determine the optimal weights of the selected assets.
- GAF-ResNet-LSTM + Equal-weighted
  Similar to the GAF-ResNet-LSTM + MV model and the proposed model, the GAF-ResNet-LSTM + Equal-Weighted approach uses the GAF-ResNet-LSTM network to forecast returns and select a subset of assets in the first stage. The difference arises in the second stage: rather than using MV or NCO optimization, it allocates capital equally across all chosen assets; that is, a naive 1/N weighting. This baseline enables us to assess whether a simple equal-weight strategy can match or outperform more sophisticated optimization methods.
- Random + NCO
  In contrast, deep learning-based selection in the proposed model, the Random + NCO strategy, uses a naive (random) procedure to select assets in the first stage. However, the number of assets chosen must match that in the GAF-ResNet-LSTM + NCO model to ensure a fair comparison. In the second stage, the NCO is again used to optimize the weights of these randomly chosen assets. This baseline helps determine whether deep learning-based asset preselection is truly beneficial compared to a purely random selection.

III. EXPERIMENT

All experiments were conducted using Google Colab on a Windows 11 system equipped with an NVIDIA GPU, an Intel i5-1235U processor (1.30 GHz), and 8 GB of system memory. The dataset contains daily OHLC price data for 25 stocks from January 1, 2014, to January 1, 2024, for a total of 2,409 trading days. Each sample is labeled according to the directional movement of the subsequent day's price.For every stock, there are rolling windows of 60 trading days. Each window is divided into three time slices to show how the market changes in the short, medium, and long term. This

Algorithm 1 Pseudo-Code for NCO Portfolio Weights

Require:
- $\lambda > 0$
- $K \in \mathbb{N}$
- n – number of assets
- $\boldsymbol{\mu} \in \mathbb{R}^n$
- $\Sigma \in \mathbb{R}^{n \times n}$

Ensure: $\mathbf{w} \in \mathbb{R}^n$

Step 1: Estimate statistics
$\boldsymbol{\mu} \leftarrow$ mean(returns, 0)
$\Sigma \leftarrow$ covariance(returns)
Step 2: Compute distance matrix
for $i = 1$ to n **do**
  for $j = 1$ to n **do**
    $\rho_{ij} \leftarrow \Sigma_{ij} / \sqrt{\Sigma_{ii}\Sigma_{jj}}$
    $d_{ij} \leftarrow \sqrt{2(1-\rho_{ij})}$
  end for
end for
Step 3: Hierarchical clustering
$D \leftarrow$ linkage(d, "ward")
$\{C_1, \ldots, C_K\} \leftarrow$ cut_tree(D, K)
Step 4: Intra-cluster optimization
for $k = 1$ to K **do**
  idx $\leftarrow C_k$
  $\boldsymbol{\mu}^{(k)} \leftarrow \boldsymbol{\mu}$[idx]
  $\Sigma^{(k)} \leftarrow \Sigma$[idx, idx]
  $\mathbf{w}^{(k)*} \leftarrow \arg\min_{\mathbf{w} \geq 0, \sum w = 1} \frac{1}{2} \mathbf{w}^\top \Sigma^{(k)} \mathbf{w}$
  Store w_intra[k], $\bar{\mu}_k$, $\bar{\sigma}_k^2$
end for
Step 5: Cluster-level covariance
for $k = 1$ to K **do**
  $\mathbf{R}[:, k] \leftarrow$ returns[:, C_k] $\times$ w_intra[k]
end for
$\bar{\Sigma} \leftarrow$ covariance($\mathbf{R}$)
Step 6: Inter-cluster optimization
$\boldsymbol{\pi}^* \leftarrow \arg\min_{\boldsymbol{\pi} \geq 0, \sum \pi = 1} \frac{1}{2} \boldsymbol{\pi}^\top \bar{\Sigma} \boldsymbol{\pi}$
Step 7: Assemble final portfolio
$\mathbf{w} \leftarrow \mathbf{0}$
for $k = 1$ to K **do**
  for all $i \in C_k$ **do**
    $w_i \leftarrow \pi_k^* \times$ w_intra[k][pos(i)]
  end for
end for
return $\mathbf{w}$

design is motivated by the multi-horizon nature of financial behavior- short slices capture local fluctuations and momentum, intermediate slices capture transitional patterns, and longer slices encode broader trends. Encoding these slices separately reduces the burden on the model to implicitly infer scale separation and improves robustness under regime shifts. Each slice is transformed into a GADF image computed on smoothed and normalized daily returns and resized to 20×20 pixels. The GADF representation preserves temporal ordering while encoding local correlations between time steps in a two-dimensional form suitable for convolutional neural networks. The choice of image resolution reflects a balance between representational fidelity and computational efficiency: increasing the resolution leads to quadratic growth in memory and computation, whereas overly small images may oversmooth informative dynamics. Empirically, $M = 20$ provides stable training and sufficient temporal detail for downstream prediction.To mitigate look-ahead bias and emulate realistic deployment conditions, we adopt a rolling walk-forward validation strategy. In each iteration, a five-year training window is followed by a six-month validation period and a subsequent six-month test period. The combined validation-test block is then advanced by six months, and the procedure is repeated until the full dataset is utilized. All preprocessing, scaling, and threshold estimation are performed strictly within the training portion of each walk. Table 2 summarizes the resulting data partitions.

A. STAGE I-PRESELECTION: PREDICTION-BASED APPROACH

Financial time-series modeling requires validation procedures that preserve temporal order, as random partitioning methods such as k-fold or leave-one-out cross-validation can introduce information leakage and violate causality. Accordingly, we employ the walk-forward validation scheme described above, ensuring that the model is always trained on historical data and evaluated exclusively on unseen future observations.

The prediction model is trained to minimize the mean squared error (MSE) of multi-horizon log-return forecasts

$$\mathcal{L} = \frac{1}{H} \sum_{k=1}^{H} \left(\hat{y}_{t,k} - y_{t,k}\right)^2$$

where H denotes the forecast horizon. Optimization is performed using the Adam optimizer with a learning rate of 5×10^{-4} and a batch size of 32. Early stopping with a patience of four epochs is applied based on validation loss. To regularize training and reduce overfitting, only the final residual block of ResNet18, the LSTM encoder, and the prediction head are updated, while earlier convolutional layers remain frozen.

To enhance interpretability, attention rollout is incorporated into the LSTM module. Attention maps are aggregated across layers and heads, yielding a hierarchical view of temporal dependencies. The resulting visualizations in Figure. 6 highlight the time steps that most strongly influence model predictions, offering simple explanations for the decision-making process and improving model transparency. The training and validation loss curves across all walk-forward folds exhibit rapid convergence and close alignment, indicating stable optimization and effective generalization of the proposed model, as presented in Figure. 7.

Performance is evaluated across five walk-forward folds for three architectures: ResNet18, LSTM, and the proposed hybrid GAF-ResNet-LSTM model. The classification

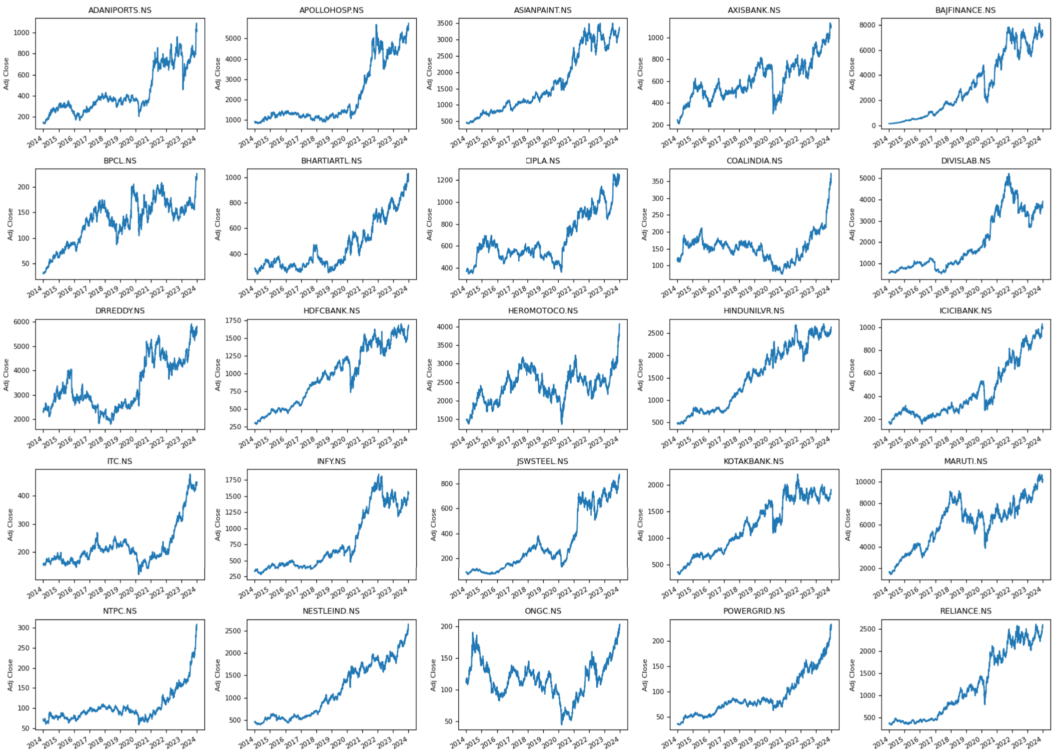

FIGURE 4. Daily adjusted close price of different stocks [45].

TABLE 2. Walk-forward validation splits.

Walk	Training Period	Validation Period	Test Period
1	2014-05-16 → 2019-05-15	2019-05-16 → 2019-11-15	2019-11-16 → 2020-05-15
2	2014-05-16 → 2020-05-15	2020-05-16 → 2020-11-15	2020-11-16 → 2021-05-15
3	2014-05-16 → 2021-05-15	2021-05-16 → 2021-11-15	2021-11-16 → 2022-05-15
4	2014-05-16 → 2022-05-15	2022-05-16 → 2022-11-15	2022-11-16 → 2023-05-15
5	2014-05-16 → 2023-05-15	2023-05-16 → 2023-11-15	2023-11-16 → 2023-12-26

TABLE 3. Average test performance.

Model	Accuracy	Recall	Precision	Macro F1	Weighted F1
GAF-ResNet	0.612	0.594	0.605	0.487	0.598
GAF-LSTM	0.624	0.607	0.618	0.503	0.611
GAF-ResNet-LSTM	0.672	0.632	0.645	0.537	0.642
Hatami et al. [19]	0.638	0.618	0.631	0.519	0.627
Barra et al. [20]	0.612	0.594	0.605	0.487	0.598
Khalid et al. [49]	0.623	0.589	0.599	0.461	0.634

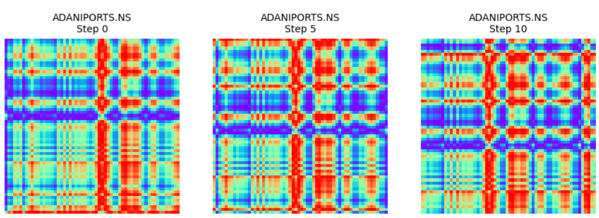

FIGURE 5. Stock images in a sequence of 3 frames converted by GAF.

performance across the three directional regimes-Down, Neutral, and Up-further validates the model's practical relevance, shown by Figure. 8. Average forecasting metrics across all test periods are reported in Table 3, while Figure. 9 illustrates representative price predictions generated by the hybrid model.

B. STAGE II-OPTIMAL PORTFOLIO CONSTRUCTION

1) ASSET CORRELATION

Figure. 10 represents a hierarchical clustering heatmap of asset correlations based on Pearson correlation and Ward linkage. The heatmap visualizes the correlation structure among different stocks, aiding in understanding their relationships for portfolio optimization. The dendrograms on the

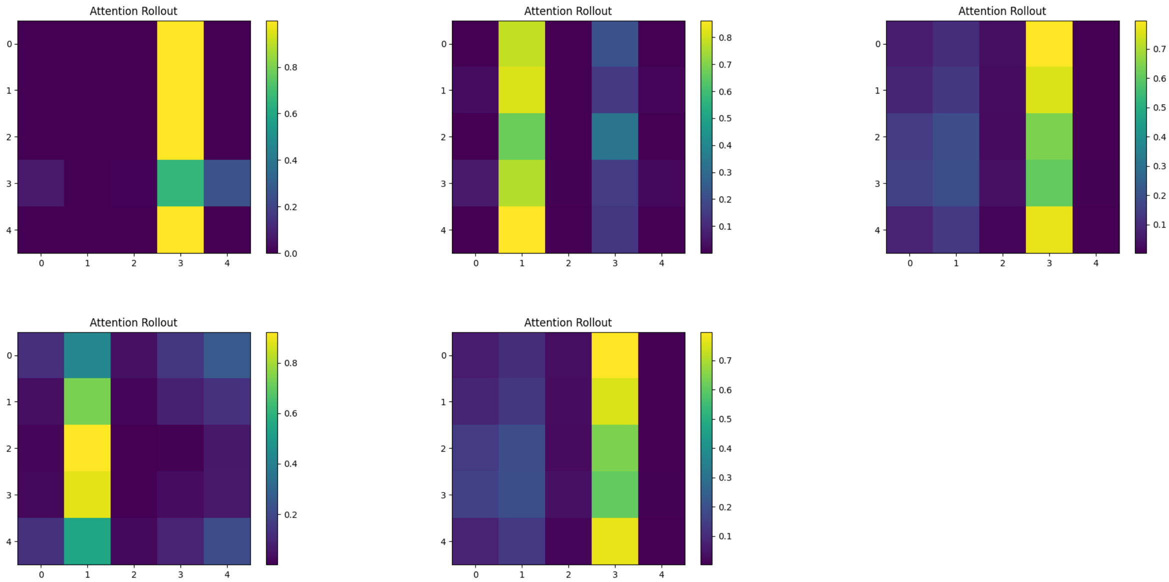

FIGURE 6. Prediction model allocates attention across input features during stock trend.

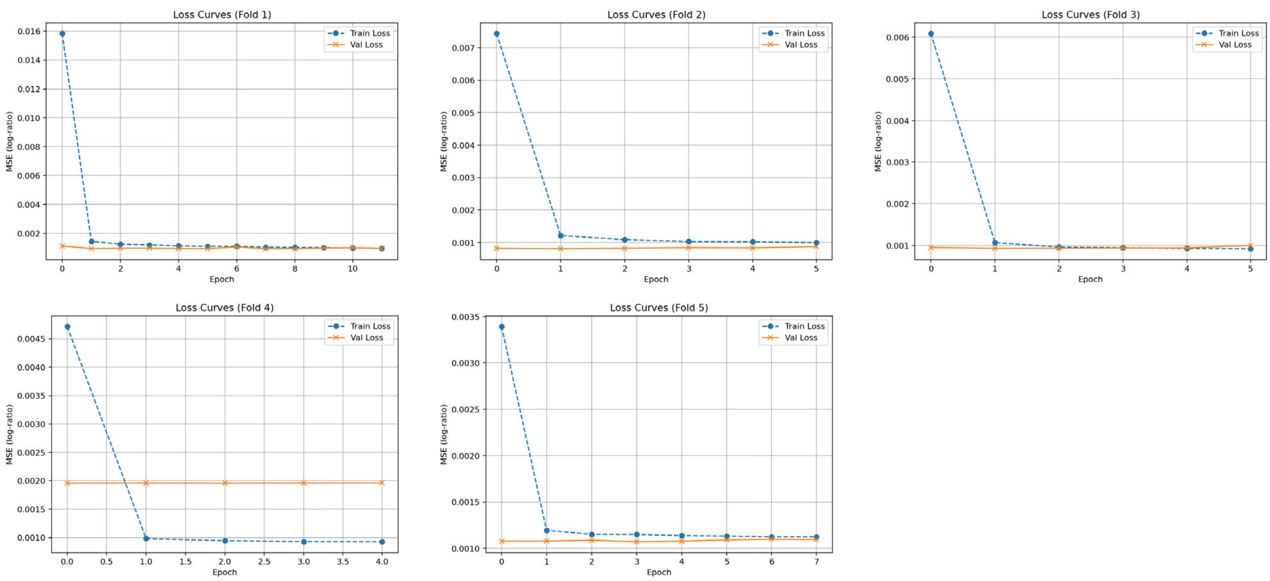

FIGURE 7. Training and validation loss curves across five folds.

top and left indicate how stocks are hierarchically grouped based on their correlations. In clustering stocks, the blue cluster and orange cluster show distinct stock groupings with strong internal correlations. The color scale ranges from −1 to 1, which are strong negative correlation and strong positive correlation, respectively. Darker blue areas indicate strong positive correlations, while lighter shades suggest weaker correlations. The magenta lines highlight significant clusters of stocks. These clusters indicate subsets of stocks that move identically, useful for portfolio diversification and risk reduction. The upper-left cluster is POWERGRID.NS, NTPC. NS, COALINDIA. NS represents highly correlated utility and energy stocks. The lower-right cluster ADANIPORTS.NS, CIPLA. NS, INFY. NS suggests a separate group from different industries. Diversifying across uncorrelated clusters can reduce risk and enhance portfolio stability.

2) OPTIMAL PORTFOLIO CONSTRUCTION

To minimize estimation errors in asset allocation, hierarchical clustering methods are utilized in the NCO model, which is chosen for portfolio optimization. The clustering method is defined using Pearson correlation, which measures linear

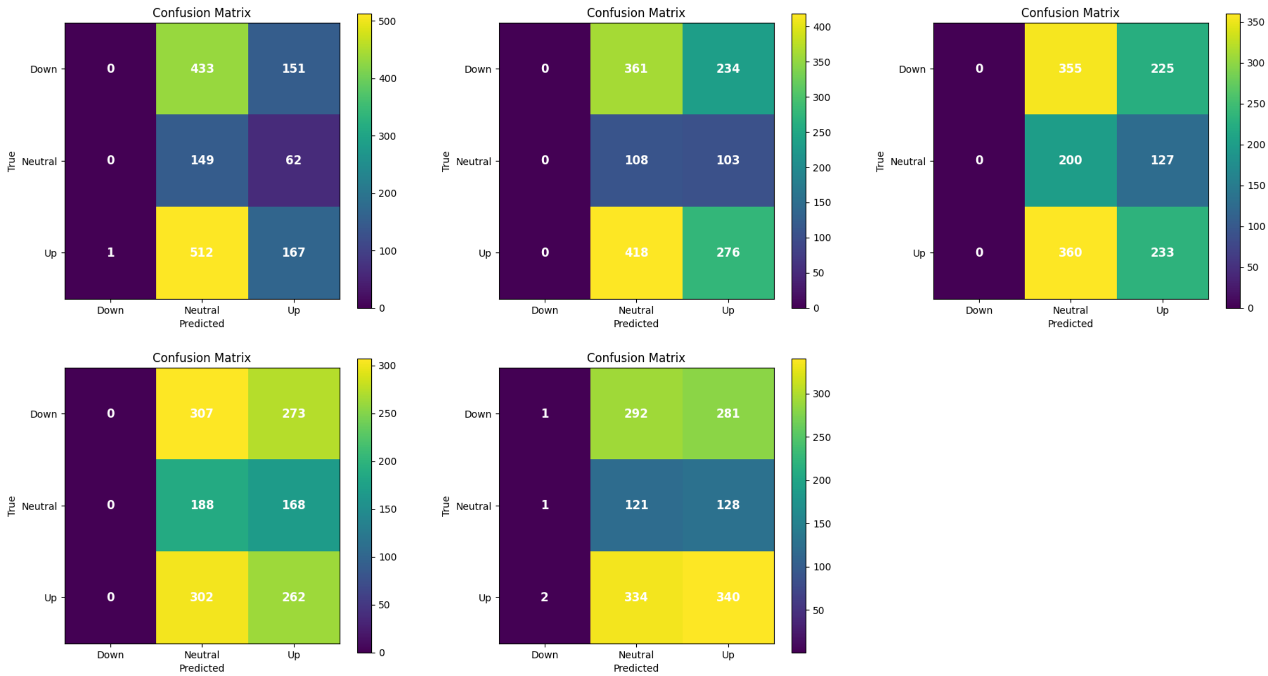

FIGURE 8. **Five folds confusion metrics across Down, Neutral, and Up trends.**

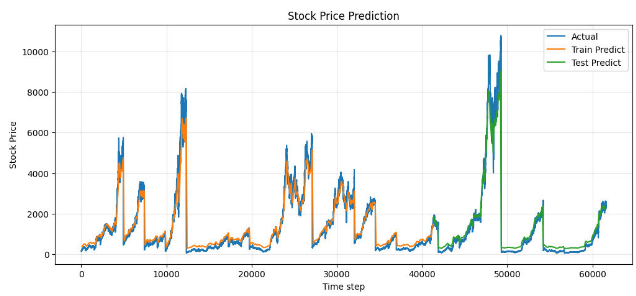

FIGURE 9. **Overall prediction performance of the GAF-ResNet-LSTM Model.**

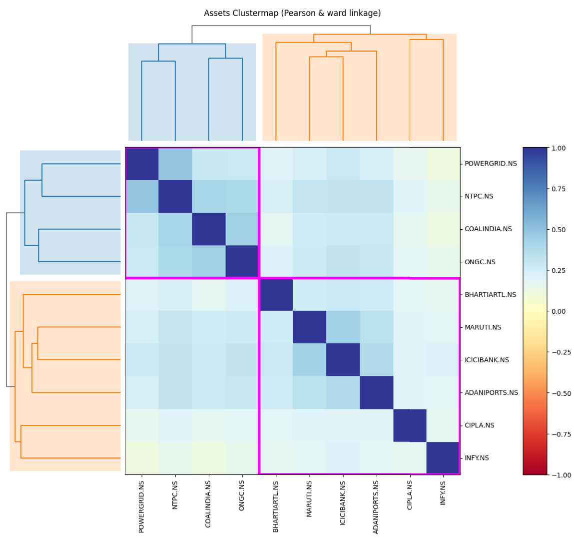

FIGURE 10. **Hierarchical clustering heatmap of asset correlations.**

relationships between assets. The optimization objective is set to MinRisk, prioritizing variance minimization. The risk is measured by mean-variance, which relies on variance for risk assessment. Ward's linkage, which minimizes intra-cluster variance, is one of the parameters for hierarchical clustering. The maximum value of k is 10, which means that there can be at most 10 clusters. The order of leaf is True, which means that Figure. 10 shows the best way to arrange assets for better cluster visualization and portfolio diversification. The structured approach enhances the robustness of portfolio allocation, reducing reliance on traditional covariance estimation and improving performance in dynamic market conditions. The average annual financial results of portfolio optimization for GAF-ResNet-LSTM+NCO are laid down in Table 5, against the baselines. Table 4 illustrates the portfolio weights assigned to ten selected stocks using three distinct allocation strategies, such as equal-weighted, NCO, and mean-variance optimization. The comparison of these methods demonstrates the impact of different optimization strategies on asset allocation. This analysis is critical in highlighting the effectiveness of predictive modeling with NCO over traditional approaches in constructing robust investment portfolios.

Figure. 12 presents a comparative analysis of portfolio performance over the period 2014 to 2024 using different portfolio optimization techniques. The y-axis represents cumulative returns, while the x-axis represents the time

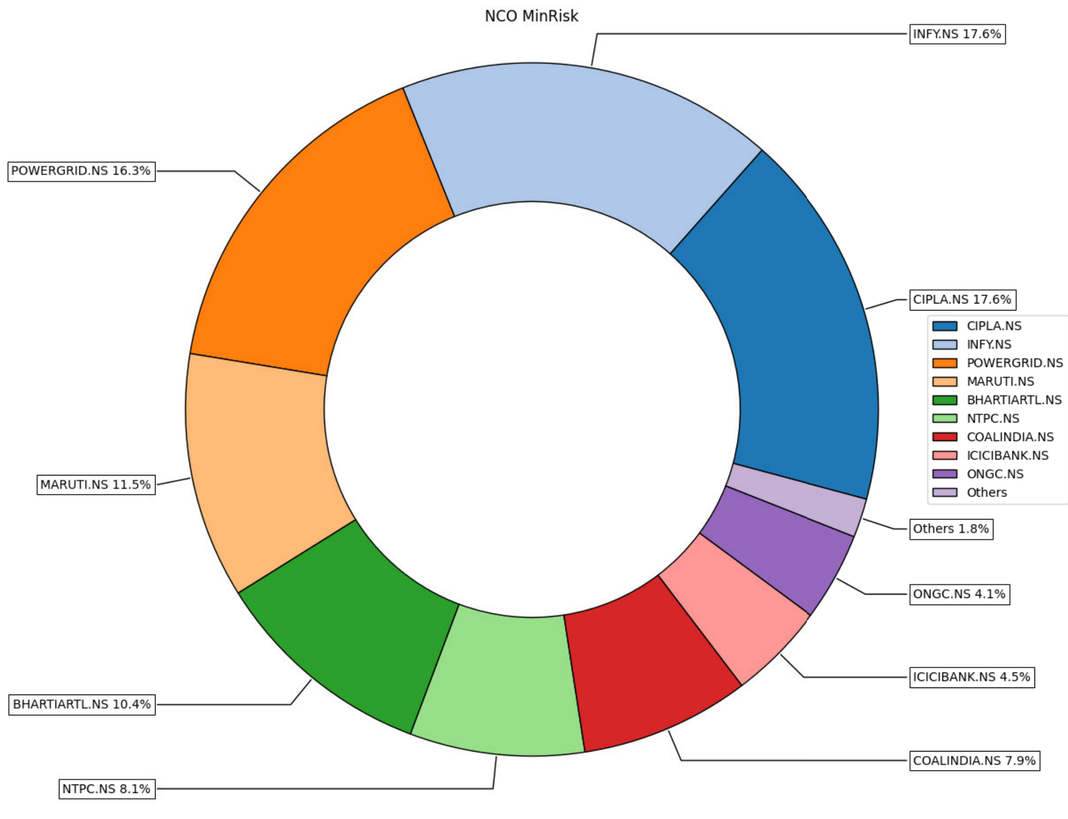

FIGURE 11. Optimal weight allocation.

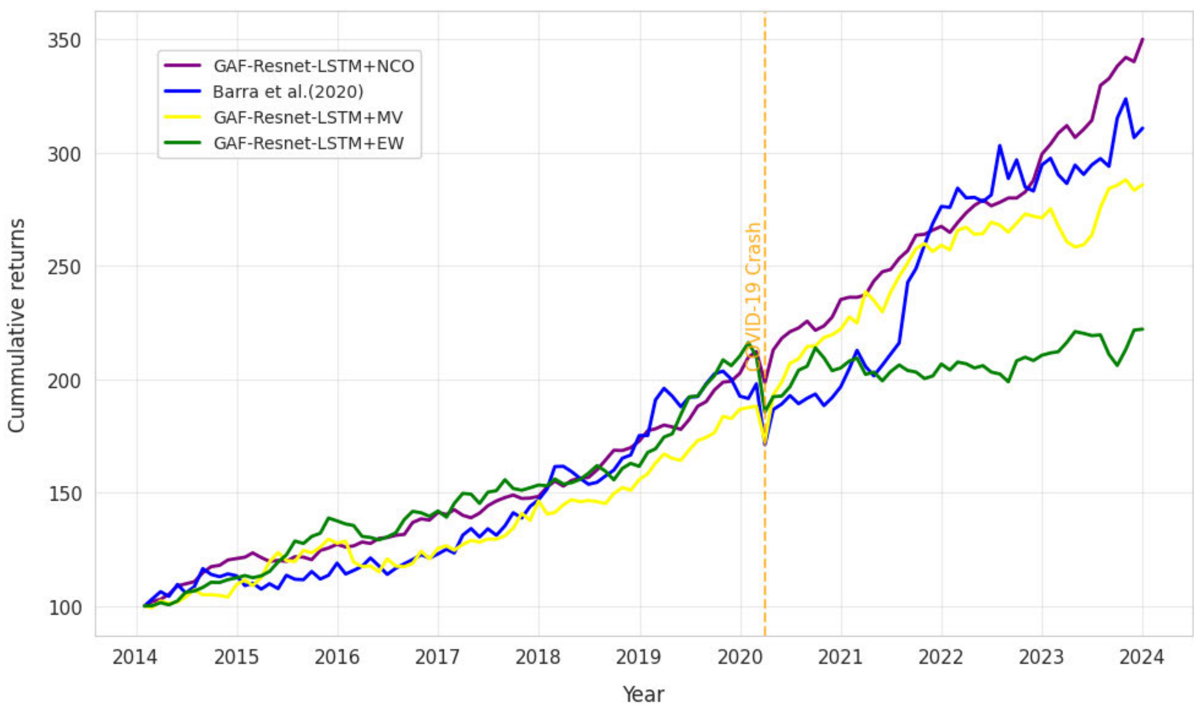

FIGURE 12. Cumulative return of different models.

period. The graph compares three deep learning-based portfolio optimization methods against a benchmark study. The proposed model shows the highest performance in terms of portfolio value growth, indicating its effectiveness in

TABLE 4. Weight allocation by different models (%).

Stocks	Equal-weighted	NCO	Mean-variance
ADANIPORTS.NS	0.1	1.8	5.12
BHARTIARTL.NS	0.1	10.4	8.42
CIPLA.NS	0.1	17.6	12.30
COALINDIA.NS	0.1	7.9	9.82
ICICIBANK.NS	0.1	4.5	6.55
INFY.NS	0.1	17.6	14.22
MARUTI.NS	0.1	11.4	10.01
NTPC.NS	0.1	8.2	7.89
ONGC.NS	0.1	4.1	5.34
POWERGRID.NS	0.1	16.3	20.25

TABLE 5. Portfolio metrics for different models.

Metrics	Benchmark index	EW	NCO	MV
Annualized Return (%)	20.92	19.11	21.05	19.29
Annualized Volatility (%)	16.37	15.46	14.8	15.45
Skewness	-0.32	-0.41	-0.33	-0.41
Kurtosis	5.48	6.67	5.37	6.82
Maximum Drawdown (%)	-18.39	-20.84	-18.2	-18.15
Sharpe Ratio	1.28	1.24	1.31	1.25

capturing financial patterns and optimizing asset allocation. The proposed approach exhibits significant outperformance, particularly from 2019 onward, demonstrating its ability to capture market trends and dynamically optimize portfolio weights. The sharp rise after 2020 suggests that NCO effectively adapts to post-pandemic market volatility. This overall shows that deep learning-based models show better adaptability to market fluctuations compared to traditional methods.

C. STATISTICAL SIGNIFICANCE

To ensure our results are statistically significant, we conduct a paired t-test [9] comparing the portfolio returns of the proposed model vs. optimized models. Let $\overline{X}_1$ and $\overline{X}_2$ be the mean returns of the proposed model and benchmark portfolios, respectively, computed over n and m observations. The test statistic is

$$t = \frac{\overline{X}_1 - \overline{X}_2}{s_p \sqrt{\frac{1}{n} + \frac{1}{m}}}$$

where the standard deviation s_p is

$$s_p = \sqrt{\frac{(n-1)\, s_{X_1}^2 + (m-1)\, s_{X_2}^2}{n+m-2}}$$

Now, to calculate the null hypothesis that the optimization of method l has no significant difference as compared to the optimization of method m, with l, $m \in$ GAF-ResNet-LSTM + Equal-Weighted, GAF-ResNet-LSTM + Mean-Variance, GAF-ResNet-LSTM + NCO, [20]. By t-test, the computed t-statistic is $t = 3.91$ with a two-sided p-value of 0.0004 ($p < 0.05$). Hence, we reject the null hypothesis H_0 of equal mean returns, confirming that the proposed hybrid model delivers a statistically significant improvement over the optimized models.

IV. DISCUSSION

Table 3 summarizes the average out-of-sample performance of three proposed architectures, GAF-ResNet, GAF-LSTM, and the hybrid GAF-ResNet-LSTM, alongside two baseline models adopted from existing literature. Across all metrics, the hybrid model attains the strongest results, reaching 0.651 accuracy, 0.632 recall, 0.645 precision, and the highest macro- and weighted-F1 scores (0.537 and 0.642, respectively). Relative to the best single-stream variant, the hybrid improves accuracy and weighted F1. When compared with the strongest literature baseline in the table, the hybrid still offers a notable margin. The classification performance across the three directional regimes-Down, Neutral, and Up-further validates the model's practical relevance. Notably, the hybrid architecture demonstrates superior capability in identifying the Down regime, a critical aspect for risk-sensitive investment strategies. While misclassifications predominantly occur between Neutral and Up classes, the Down class remains distinctly recognized, suggesting that the model effectively captures bearish market signals. The average test accuracy reaches 67.2%, representing a 10-11% improvement over benchmark models. The macro-F1 scores reflect balanced classification across regimes, reinforcing the model's generalizability. Confusion matrices further highlight that most misclassifications occur between *Neutral* and *Up* classes, while the *Down* class is robustly distinguished, as presented in Figure. 8.

Ablation studies provide deeper insight into the architectural contributions. The removal of the LSTM component results in a 3.5% drop in accuracy, highlighting its role in capturing temporal dependencies across GAF image sequences. Additionally, excluding multi-resolution imaging adversely affects recall and precision in the Down class, indicating that both fine-and coarse-grained perspectives are essential for nuanced regime detection. These findings advocate for the joint use of convolutional and recurrent structures in financial time series modeling.

Table 5 compares four portfolio construction methods along standard risk-return diagnostics. The NCO portfolio delivers the highest annualized return (21.05%) and the highest sharpe ratio (1.31) while also exhibiting the lowest volatility (14.8%) among the tested methods. Relative to the benchmark index, NCO improves the return by 13 basis points and lifts risk-adjusted performance despite a 1.6 percentage point reduction in volatility. Against traditional mean-variance allocation, NCO increases return by 1.76 percentage points and raises sharpe from 1.25 to 1.31, suggesting that clustering assets prior to optimization mitigates estimation error and improves diversification compared with a Markowitz solution. The daily cumulative return for each model is shown in Figure. 12. The average annualized cumulative returns of the baseline models GAF-ResNet-LSTM + MV, GAF-ResNet-LSTM + EW, and Random +

NCO are 19.29%, 19.11%, and 14.34%, respectively, while the average annualized cumulative return of the benchmark index is 20.92%.

V. CONCLUSION AND FUTURE WORK

A. CONCLUSION

This study proposed a novel structure for asset selection by combining GAF representations with a hybrid ResNet-LSTM model, alongside a nested clustered optimization for portfolio construction. The GAF transformation retains temporal patterns while converting sequential financial data into two-dimensional formats suitable for convolutional learning. Within this model, the ResNet component extracts layered spatial features, while the LSTM captures extended temporal relationships, together offering a detailed view of market behavior.Training employs walk-forward validation and a composite loss combining mean squared error and directional accuracy to balance calibration and classification, reaching 0.651 accuracy and the highest macro- and weighted-F1 scores, 0.537 and 0.642, respectively. Upon coupling these forecasts with the NCO algorithm, where Pearson-Ward clustering groups economically coherent assets before constraint-driven allocation, the model delivers superior out-of-sample performance over the 2014-2024 period. Specifically, it achieves the highest cumulative returns compared to classical mean-variance and equal-weighted strategies, as well as a recent deep learning-based benchmark of [20], illustrated in Figure. 12. In Table 5, the proposed framework also attains the highest Sharpe, the lowest maximum drawdown (−18.2 %), and maintains competitive annualized volatility (14.8 %). The findings highlight the benefit of integrating forecasting techniques within a structured optimization pipeline. This leads to improved prediction accuracy and better risk-adjusted returns in equity portfolio management.

B. LIMITATIONS AND FUTURE WORK

Despite the strong empirical performance of the proposed GAF-ResNet-LSTM model combined with nested clustered optimization, several limitations should be acknowledged. First, the current framework primarily relies on historical price-based information for feature construction. Although the GAF representation effectively captures nonlinear temporal patterns, it does not explicitly incorporate other informative market signals such as trading volume, corporate actions, macroeconomic announcements, or unstructured data sources such as financial news and sentiment. Integrating these complementary information channels may further enhance predictive robustness and improve portfolio allocation decisions, particularly during regime shifts driven by exogenous events. Second, while the hybrid deep learning architecture demonstrates strong predictive and allocative performance, it introduces additional computational complexity compared to traditional portfolio optimization methods. Although training is feasible on modern hardware, future work may explore model compression, parameter sharing, or lightweight architectures to improve scalability and facilitate real-time deployment.

Future work may focus on extending this framework to augment the predictor with volume, corporate events, macroeconomic announcements, and news sentiment signals and evaluate alternative time-series image encodings. Exploring reinforcement learning to support dynamic trading strategies. Overall, the proposed GAF-ResNet-LSTM with nested clustered optimization represents a significant step forward in the design of interpretable and resilient deep learning models for asset selection and portfolio optimization.

ACKNOWLEDGMENT

Neha Solanki acknowledges MHRD, New Delhi, Government of India. She also thankful to Priya Singh for her advise and support during the preparation of the manuscript.

REFERENCES

[1] P. Gupta, M. K. Mehlawat, and A. Z. Khan, "Multi-period portfolio optimization using coherent fuzzy numbers in a credibilistic environment," *Expert Syst. Appl.*, vol. 167, pp. 114–135, 2021.
[2] J. K. Pahade and M. Jha, "Credibilistic variance and skewness of trapezoidal fuzzy variable and mean–variance–skewness model for portfolio selection," *Results Appl. Math.*, vol. 11, Apr. 2021, Art. no. 100159.
[3] H. M. Markowitz, "Portfolio selection," *J. Finance*, vol. 7, no. 1, p. 77, Mar. 1952.
[4] T. Fischer and C. Krauß, "Deep learning with long short-term memory networks for financial market predictions," *Eur. J. Oper. Res.*, vol. 270, no. 2, pp. 654–669, Dec. 2018.
[5] J. B. Guerard, H. M. Markowitz, and G. Xu, "Earnings forecasting in a global stock selection model and efficient portfolio construction and management," *Int. J. Forecast.*, vol. 31, no. 2, pp. 550–560, Jan. 2015.
[6] C. N. Babu and B. E. Reddy, "Selected Indian stock predictions using a hybrid ARIMA-GARCH model," in *Proc. Int. Conf. Adv. Electron. Comput. Commun.*, Bangalore, India, Oct. 2014, pp. 1–6.
[7] P. Mondal, L. Shit, and S. Goswami, "Study of effectiveness of time series modeling (Arima) in forecasting stock prices," *Int. J. Comput. Sci. Eng. Appl.*, vol. 4, no. 2, pp. 13–29, Apr. 2014.
[8] A. M. Rather, A. Agarwal, and V. N. Sastry, "Recurrent neural network and a hybrid model for prediction of stock returns," *Expert Syst. Appl.*, vol. 42, no. 6, pp. 3234–3241, Dec. 2015.
[9] T. K. Kim, "T test as a parametric statistic," *Korean J. Anesthesiol.*, vol. 68, no. 6, p. 540, Jan. 2015.
[10] Z. Jiang, D. Xu, and J. Liang, "A deep reinforcement learning framework for the financial portfolio management problem," 2017, *arXiv:1706.10059*.
[11] B. S. Dees, L. Stankovic, A. G. Constantinides, and D. P. Mandic, "Portfolio cuts: A graph-theoretic framework to diversification," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Barcelona, Spain, May 2020, pp. 8454–8458.
[12] Y. Ma, W. Wang, and Q. Ma, "A novel prediction based portfolio optimization model using deep learning," *Comput. Ind. Eng.*, vol. 177, Jan. 2023, Art. no. 109023.
[13] F. D. Paiva, R. T. N. Cardoso, G. P. Hanaoka, and W. M. Duarte, "Decision-making for financial trading: A fusion approach of machine learning and portfolio selection," *Expert Syst. Appl.*, vol. 115, pp. 635–655, Aug. 2018.
[14] W. Wang, W. Li, N. Zhang, and K. Liu, "Portfolio formation with preselection using deep learning from long-term financial data," *Expert Syst. Appl.*, vol. 143, Oct. 2020, Art. no. 113042.
[15] M. Lee, J. Liang, Y. Wu, G. Chiou, C.-Y. Hsu, C.-Y. Wang, J.-W. Lin, and C. Tsai, "High school students' conceptions of science laboratory learning, perceptions of the science laboratory environment, and academic self-efficacy in science learning," *Int. J. Sci. Math. Educ.*, vol. 18, no. 1, pp. 1–18, Feb. 2020.
[16] Y. Ma, R. Han, and W. Wang, "Portfolio optimization with return prediction using deep learning and machine learning," *Expert Syst. Appl.*, vol. 165, Sep. 2021, Art. no. 113973.

[17] Y. Ma, R. Han, and X. Fu, "Stock prediction based on random forest and LSTM neural network," in *Proc. 19th Int. Conf. Control, Autom. Syst. (ICCAS)*, Jeju, South Korea, Oct. 2019, pp. 126–130.
[18] Z. Wang and T. Oates, "Imaging time-series to improve classification and imputation," 2015, *arXiv:1506.00327*.
[19] N. Hatami, Y. Gavet, and J. Debayle, "Classification of time-series images using deep convolutional neural networks," in *Proc. 10th Int. Conf. Mach. Vis. (ICMV)*, vol. 10696, Vienna, Austria, Nov. 2017, pp. 242–249.
[20] S. Barra, S. M. Carta, A. Corriga, A. S. Podda, and D. R. Recupero, "Deep learning and time series-to-image encoding for financial forecasting," *IEEE/CAA J. Autom. Sinica*, vol. 7, no. 3, pp. 683–692, May 2020.
[21] R. Akita, A. Yoshihara, T. Matsubara, and K. Uehara, "Deep learning for stock prediction using numerical and textual information," in *Proc. IEEE/ACIS 15th Int. Conf. Comput. Inf. Sci. (ICIS)*, Okayama, Japan, Jun. 2016, pp. 1–6.
[22] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Las Vegas, NV, USA, Jun. 2016, pp. 770–778.
[23] Ö. B. Sezer and A. M. Özbayoğlu, "Algorithmic financial trading with deep convolutional neural networks: Time series to image conversion approach," *Appl. Soft Comput.*, vol. 70, pp. 525–538, Apr. 2018.
[24] Y. Komori, "Convolutional neural network for stock price prediction using transfer learning," Social Sci. Res. Netw., Rochester, NY, USA, Tech. Rep. ssrn.3756702, Dec. 2020.
[25] T. Kim and H. Y. Kim, "Forecasting stock prices with a feature fusion LSTM-CNN model using different representations of the same data," *PLoS ONE*, vol. 14, no. 2, Feb. 2019, Art. no. e0212320.
[26] S. W. Lee and H. Y. Kim, "Stock market forecasting with super-high dimensional time-series data using ConvLSTM, trend sampling, and specialized data augmentation," *Expert Syst. Appl.*, vol. 161, Jul. 2020, Art. no. 113704.
[27] G. Costa and R. H. Kwon, "A robust framework for risk parity portfolios," *J. Asset Manag.*, vol. 21, no. 5, pp. 447–466, Aug. 2020.
[28] M. M. L. de Prado, *Machine Learning for Asset Managers*. Cambridge, U.K.: Cambridge Univ. Press, 2020.
[29] K. Imajo, K. Minami, K. Ito, and K. Nakagawa, "Deep portfolio optimization via distributional prediction of residual factors," in *Proc. AAAI Conf. Artif. Intell.*, vol. 35, May 2021, pp. 213–222.
[30] P. Khodaee, A. Esfahanipour, and H. Taheri, "Forecasting turning points in stock price by applying a novel hybrid CNN-LSTM-ResNet model fed by 2D segmented images," *Eng. Appl. Artif. Intell.*, vol. 116, Oct. 2022, Art. no. 105464.
[31] J. Du, "Mean–variance portfolio optimization with deep learning based-forecasts for cointegrated stocks," *Expert Syst. Appl.*, vol. 201, Apr. 2022, Art. no. 117005.
[32] P. Singh, M. Jha, M. Sharaf, M. A. El-Meligy, and T. R. Gadekallu, "Harnessing a hybrid CNN-LSTM model for portfolio performance: A case study on stock selection and optimization," *IEEE Access*, vol. 11, pp. 104000–104015, 2023.
[33] P. Singh and M. Jha, "Portfolio optimization using novel EW-MV method in conjunction with asset preselection," *Comput. Econ.*, vol. 64, no. 6, pp. 3683–3712, Mar. 2024.
[34] P. Peng, Y. Chen, W. Lin, and J. Z. Wang, "Attention-based CNN–LSTM for high-frequency multiple cryptocurrency trend prediction," *Expert Syst. Appl.*, vol. 237, Sep. 2023, Art. no. 121520.
[35] J. González-Bueno, R. Tamošiūnien, C. G. Morales, and G. Rueda-Barrios, "Applying the mean-variance framework: Portfolio optimization and comparative performance analysis in the emerging Colombian capital market," *Bus. Manage. Econ. Eng.*, vol. 23, no. 1, pp. 164–188, Jun. 2025.
[36] J. Gonzalez-Bueno, F. Garcia, J. L. Garcés-Bautista, J. Oliver, and R. Tamošiūnien, "A multi-objective credibilistic approach to portfolio optimization: Balancing return, risk, and liquidity in emerging and developed markets," *Intelekt. Ekonom.*, vol. 18, no. 2, pp. 284–307, Jan. 2023.
[37] X. Huang and D. Ma, "Uncertain mean-chance model for portfolio selection with multiplicative background risk," *Int. J. Syst. Sci., Oper. Logist.*, vol. 10, no. 1, Dec. 2022, Art. no. 2158443.
[38] X. Wang, B. Wang, T. Li, H. Li, and J. Watada, "Multi-criteria fuzzy portfolio selection based on three-way decisions and cumulative prospect theory," *Appl. Soft Comput.*, vol. 134, Jan. 2023, Art. no. 110033.
[39] M. L. de Prado, "A robust estimator of the efficient frontier," *SSRN Electron. J.*, 2016, Art. no. 3469961.
[40] F. Espiga-Fernández, Á. G. Sánchez, and J. Ordieres-Meré, "A systematic approach to portfolio optimization: A comparative study of reinforcement learning agents, market signals, and investment horizons," *Algorithms*, vol. 17, no. 12, p. 570, Dec. 2024.
[41] V. Lemieux, P. S. Rahmdel, R. Walker, B. L. W. Wong, and M. Flood, "Clustering techniques and their effect on portfolio formation and risk analysis," in *Proc. Int. Workshop Data Sci. Macro-Model.*, 2014, pp. 1–6.
[42] A. Sharma, C. K. Verma, and P. Singh, "Enhancing option pricing accuracy in the Indian market: A CNN-BiLSTM approach," *Comput. Econ.*, vol. 65, no. 6, pp. 3751–3778, Aug. 2024.
[43] W. Hu and A. N. Kercheval, "Portfolio optimization for student-T and skewed-T returns," *Quantum Finance*, vol. 10, no. 1, pp. 91–105, 2010.
[44] W. Zhang, S. Li, Z. Guo, and Y. Yang, "A hybrid forecasting model based on deep learning feature extraction and statistical arbitrage methods for stock trading strategies," *J. Forecast.*, vol. 42, no. 7, pp. 1729–1749, 2023.
[45] N. Solanki and M. Jha, "A decade of stock data: Predictive modelling with hybrid CNN-BiLSTM technique," in *Proc. IEEE Silchar Subsection Conf. (SILCON)*, Silchar, India, Nov. 2024, pp. 1–6.
[46] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, Nov. 1997.
[47] F. A. Gers, J. Schmidhuber, and F. Cummins, "Learning to forget: Continual prediction with LSTM," *Neural Comput.*, vol. 12, no. 10, pp. 2451–2471, Oct. 2000.
[48] K. Greff, R. K. Srivastava, J. Koutník, B. R. Steunebrink, and J. Schmidhuber, "LSTM: A search space Odyssey," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 28, no. 10, pp. 2222–2232, Oct. 2017.
[49] K. Tatane, S. M. Rida, and Z. J. Taher, "From time series to images: Revolutionizing stock market predictions with convolutional deep neural networks," *Int. J. Adv. Comput. Sci. Appl.*, vol. 15, no. 1, Jan. 2024.

NEHA SOLANKI received the B.Sc. and M.Sc. degrees in mathematics from the Government Holkar Science College, Indore, Madhya Pradesh, India. She is currently an Assistant Professor with the Department of Mathematics, Government Nirbhay Singh College, Indore. She continues the Doctoral Research with the Department of Mathematics, Bioinformatics, and Computer Applications, Maulana Azad National Institute of Technology, Bhopal, Madhya Pradesh. Her research interests include machine learning approaches for asset selection and portfolio construction.

MANOJ JHA received the B.Sc. and M.Sc. degrees in mathematics from Barkatullah University, Bhopal, Madhya Pradesh, India, the M.Phil. degree from DAVV, Indore, Madhya Pradesh, and the Ph.D. degree in financial mathematics from the Maulana Azad National Institute of Technology (MANIT), Bhopal. He is currently an Associate Professor with the Department of Mathematics, Bioinformatics, and Computer Applications, MANIT. He authored three books with various research publications. His research interests include financial mathematics and portfolio optimization.